

# IELTS

## Workbook



# Reading

# 5

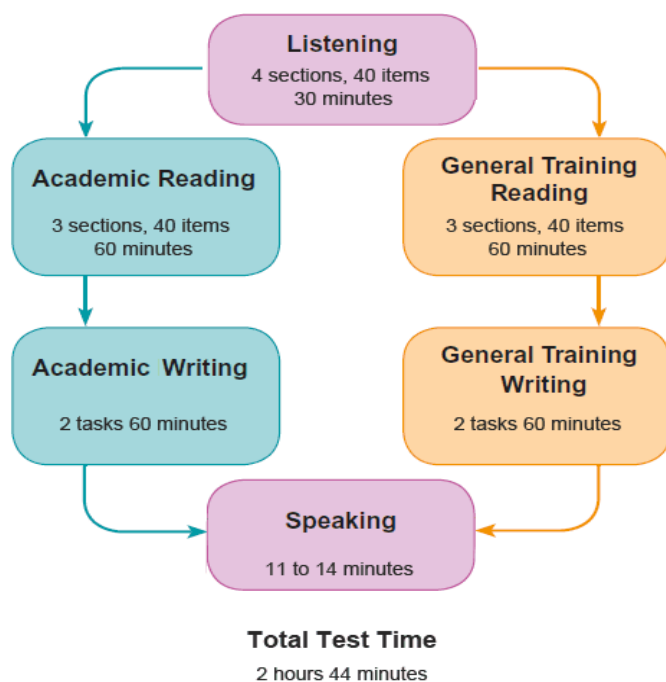
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## Preface

IELTS (International English Language Testing System) is a worldwide recognized official English exam. The IELTS is administered by the British Council, IDP: IELTS Australia and Cambridge English Language Assessment. Its aim is to evaluate the English level of the people that take it. This is one of the most accepted English exams by higher education institutions to certify English language proficiency.



**IELTS Reading:** Academic Module – 3 long texts of an academic nature. General Training Module - Mixture of long and short texts of a more general nature, as well as texts related to work situation.

### Reading Paper is Assessed on:

- Identifying the overall purpose
- Follow key arguments in a text
- Identify opinions and attitudes
- Locate specific information
- Distinguish main ideas from details.
- Extract information from a text
- Complete a diagram, summary
- Edit a table or set of notes.

### Reading Types of Questions:

- Matching Heading Questions
- True/ False/Not Given or Yes / No/ Not Given.
- Matching Paragraph Information Question.
- Summary Completion Question
- Sentence Completion Question
- **Multiple Choice Question**
- List Selection
- Choosing a Title
- Categorization Question
- Matching Sentence Endings
- Table Completion
- Flow chart Completion Question
- Short Answer Question

Multiple choice questions are usually time-consuming because the options given are most of the time very confusing. Choosing the correct option becomes difficult as it appears as though all options are related to the question asked. On an average 7-8 questions are asked in each reading test. These questions are of 2 kinds, one wherein you need to select one answer from the given 3-4 alternatives and second where you need to choose 2 answers from the given 5 alternatives. So let's understand the important guideline to strategies to solve these kinds of questions.

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## **TO SOLVE MULTIPLE CHOICE QUESTIONS:**

Multiple choice questions are usually time-consuming because the options given are most of the time very confusing. Choosing the correct option becomes difficult as it appears as though all options are related to the question asked. On an average 7-8 questions are asked in each reading test. These questions are of 2 kinds, one wherein you need to select one answer from the given 3-4 alternatives and second where you need to choose 2 answers from the given 5 alternatives. Carefully read what is mentioned in the question to avoid confusion.

### **GUIDELINES TO SOLVE:**

1. First skim read the entire passage to get the overall idea of the passage.
2. Give headings to the paragraphs for self-understanding while reading.
3. Go through the multiple-choice questions. Read the first one, find out exactly what details you're looking for (highlight those words) and try locating the paragraph containing the above information.
4. For such questions you need to read more than 1 sentence in the paragraph.
5. Accordingly choose the option which best suits the question asked.
6. Or else follow the **Mantra of Elimination**: eliminate the options which are not correct as per you and do not suit the details which you're expected to find out. Here you need to be very sure while choosing the options to be rejected.
7. Ultimately, you're left with fewer options which give you more clarity on the answer.

**Let us look at the application of the above-mentioned guidelines for the following passage below.**

## Solved Reading Passage

### THE MEANING AND POWER OF SMELL

*The sense of smell, or olfaction, is powerful. Odours affect us on a physical, psychological and social level. For the most part, however, we breathe in the aromas which surround us without being consciously aware of their importance to us. It is only when the faculty of smell is impaired for some reason that we begin to realise the essential role the sense of smell plays in our sense of well-being*

- A. A survey conducted by Anthony Synott at Montreal's Concordia University asked participants to comment on how important smell was to them in their lives. It became apparent that smell can evoke strong emotional responses. A scent associated with a good experience can bring a rush of joy, while a foul odour or one associated with a bad memory may make us grimace with disgust. Respondents to the survey noted that many of their olfactory likes and dislikes were based on emotional associations. Such associations can be powerful enough so that odours that we would generally label unpleasant become agreeable, and those that we would generally consider fragrant become disagreeable for particular individuals. The perception of smell, therefore, consists not only of the sensation of the odours themselves, but of the experiences and emotions associated with them.
- B. Odours are also essential cues in social bonding. One respondent to the survey believed that there is no true emotional bonding without touching and smelling a loved one. In fact, infants recognise the odours of their mothers soon after birth and adults can often identify their children or spouses by scent. In one well-known test, women and men were able to distinguish by smell alone clothing worn by their marriage partners from similar clothing worn by other people. Most of the subjects would probably never have given much thought to odour as a cue for identifying family members before being involved in the test, but as the experiment revealed, even when not consciously considered, smells register.
- C. In spite of its importance to our emotional and sensory lives, smell is probably the most undervalued sense in many cultures. The reason often given for the low regard in which smell is held is that, in comparison with its importance among animals, the human sense of smell is feeble and undeveloped. While it is true that the olfactory powers of humans are nothing like as fine as those possessed by certain animals, they are still remarkably acute. Our noses are able to recognise thousands of smells, and to perceive odours which are present only in extremely small quantities.
- D. Smell, however, is a highly elusive phenomenon. Odours, unlike colours, for instance, cannot be named in many languages because the specific vocabulary simply doesn't exist. 'It smells like . . . ,' we have to say when describing an odour, struggling to express our olfactory experience. Nor can odours be recorded: there is no effective way to either capture or store them over time. In the realm of olfaction, we must make do with descriptions and recollections. This has implications for olfactory research.

- E. Most of the research on smell undertaken to date has been of a physical scientific nature. Significant advances have been made in the understanding of the biological and chemical nature of olfaction, but many fundamental questions have yet to be answered. Researchers have still to decide whether smell is one sense or two - one responding to odours proper and the other registering odourless chemicals in the air. Other unanswered questions are whether the nose is the only part of the body affected by odours, and how smells can be measured objectively given the nonphysical components. Questions like these mean that interest in the psychology of smell is inevitably set to play an increasingly important role for researchers.
- F. However, smell is not simply a biological and psychological phenomenon. Smell is cultural, hence it is a social and historical phenomenon. Odours are invested with cultural values: smells that are considered to be offensive in some cultures may be perfectly acceptable in others. Therefore, our sense of smell is a means of, and model for, interacting with the world. Different smells can provide us with intimate and emotionally charged experiences and the value that we attach to these experiences is interiorised by the members of society in a deeply personal way. Importantly, our commonly held feelings about smells can help distinguish us from other cultures. The study of the cultural history of smell is, therefore, in a very real sense, an investigation into the essence of human culture.

**Choose the correct letter A, B, C and D**

1. According to the introduction, we become aware of the importance of smell when
  - A. We discover new smell.
  - B. We experience a powerful smell
  - C. Our ability to smell is damaged.
  - D. We are surrounded by odours.
  
2. The experiment described in paragraph B
  - A. Shows how we make use of smell without realising it.
  - B. Demonstrates the family members have a similar smell.
  - C. Proves that a sense of smell is learnt.
  - D. Compares the sense of smell in males and females.
  
3. What is the writer doing in paragraph C?
  - A. Supporting other research
  - B. Making a proposal
  - C. Rejecting a common belief
  - D. Describing limitation

4. What does the writer suggest about the study of smell in the atmosphere in Paragraph E?
- A. The measurement of smell is becoming more accurate.
  - B. Researchers believe smell is purely physical reaction.
  - C. Most smells are inoffensive
  - D. Smell is yet to be defined.

### **Solved Examples:**

1. According to the introduction, we become aware of the importance of smell when

Here the underlined words give an idea of what information we are looking for. It's about when we realise the importance of smell.

These details are present in the subtitle. "It is only when the faculty of smell is impaired for some reason..... plays in our sense of well being." This sentence relates to option C which says: Our ability to smell is damaged. The information given in other options are mentioned in subtitle but none are related to awareness of the importance of smell. So the correct **option is C**.

2. The experiment described in paragraph B

The question itself tells us that the information is asked from paragraph B. It is related to the experiment described. Here we use the **mantra of elimination**. **Option B** says it demonstrates that family members have similar smell. There's no description about the similarity of smell among family members. **Option C** says it proves that a sense of smell is learnt. The sense of smell is naturally present we only realise it when the organ to detect smell is injured. **Option D** says it compares the sense of smell in males and females. Here the experiment was carried out for spouses and not for men and women as an individual. But if we see option A, it says how we make use of smell without realising it. This appears similar in meaning to the last line of the paragraph, '.....but as the experiment revealed even when not consciously considered, smells register.' So, the correct answer is **option A**

## **Practise Reading Passage 1.1**

### **NATURE OR NURTURE?**

- A. A few years ago, in one of the most fascinating and disturbing experiments in behavioural psychology, Stanley Milgram of Yale University tested 40 subjects from all walks of life for their willingness to obey instructions given by a 'leader' in a situation in which the subjects might feel a personal distaste for the actions they were called upon to perform. Specifically, Milgram told each volunteer 'teacher-subject' that the experiment was in the noble cause of education, and was designed to test whether or not punishing pupils for their mistakes would have a positive effect on the pupils' ability to learn.
- B. Milgram's experimental set-up involved placing the teacher-subject before a panel of thirty switches with labels ranging from '15 volts of electricity (slight shock)' to '450 volts (danger - severe shock)' in steps of 15 volts each. The teacher-subject was told that whenever the pupil gave the wrong answer to a question, a shock was to be administered, beginning at the lowest level and increasing in severity with each successive wrong answer. The supposed 'pupil' was in reality an actor hired by Milgram to simulate receiving the shocks by emitting a spectrum of groans, screams and writhings together with an assortment of statements and expletives denouncing both the experiment and the experimenter. Milgram told the teacher-subject to ignore the reactions of the pupil, and to administer whatever level of shock was called for as per the rule governing the experimental situation of the moment.
- C. As the experiment unfolded, the pupil would deliberately give the wrong answers to questions posed by the teacher, thereby bringing on various electrical punishments, even up to the danger level of 300 volts and beyond. Many of the teacher-subjects balked at administering the higher levels of punishment, and turned to Milgram with questioning looks and/or complaints about continuing the experiment. In these situations, Milgram calmly explained that the teacher-subject was to ignore the pupil's cries for mercy and carry on with the experiment. If the subject was still reluctant to proceed, Milgram said that it was important for the sake of the experiment that the procedure be followed through to the end. His final argument was, 'You have no other choice. You must go on.' What Milgram was trying to discover was the number of teacher-subjects who would be willing to administer the highest levels of shock, even in the face of strong personal and moral revulsion against the rules and conditions of the experiment.
- D. Prior to carrying out the experiment, Milgram explained his idea to a group of 39 psychiatrists and asked them to predict the average percentage of people in an ordinary population who would be willing to administer the highest shock level of 450 volts. The overwhelming consensus was that virtually all the teacher-subjects would refuse to obey the experimenter. The psychiatrists felt that 'most subjects would not go beyond 150 volts' and they further anticipated that only four per cent would go up to 300 volts. Furthermore, they

thought that only a lunatic fringe of about one in 1.000 would give the highest shock of 450 volts.

- E. What were the actual results? Well, over 60 per cent of the teacher-subjects continued to obey Milgram up to the 450-volt limit! In repetitions of the experiment in other countries, the percentage of obedient teacher-subjects was even higher, reaching 85 per cent in one country. How can we possibly account for this vast discrepancy between what calm, rational, knowledgeable people predict in the comfort of their study and what pressured, flustered, but cooperative teachers' actually do in the laboratory of real life?
- F. One's first inclination might be to argue that there must be some sort of built-in animal aggression instinct that was activated by the experiment, and that Milgram's teacher-subjects were just following a genetic need to discharge this pent-up primal urge onto the pupil by administering the electrical shock. A modern hard-core sociobiologist might even go so far as to claim that this aggressive instinct evolved as an advantageous trait, having been of survival value to our ancestors in their struggle against the hardships of life on the plains and in the caves, ultimately finding its way into our genetic make-up as a remnant of our ancient animal ways.
- G. An alternative to this notion of genetic programming is to see the teacher-subjects' actions as a result of the social environment under which the experiment was carried out. As Milgram himself pointed out. Most subjects in the experiment see their behaviour in a larger context that is benevolent and useful to society - the pursuit of scientific truth. The psychological laboratory has a strong claim to legitimacy and evokes trust and confidence in those who perform there. An action such as shocking a victim, which in isolation appears evil, acquires a completely different meaning when placed in this setting'.
- H. Thus, in this explanation the subject merges his unique personality and personal and moral code with that of larger institutional structures, surrendering individual properties like loyalty, self-sacrifice and discipline to the service of malevolent systems of authority.
- I. Here we have two radically different explanations for why so many teacher-subjects were willing to forgo their sense of personal responsibility for the sake of an institutional authority figure. The problem for biologists, psychologists and anthropologists is to sort out which of these two polar explanations is more plausible. This, in essence, is the problem of modern sociobiology - to discover the degree to which hard-wired genetic programming dictates, or at least strongly biases, the interaction of animals and humans with their environment, that is, their behaviour. Put another way sociobiology is concerned with elucidating the biological basis of all behaviour.

**Choose the correct letter, A, B, C or D.** Write your answers next to the question number.

1. The teacher-subjects were told that they were testing whether
  - A. A 450-volt shock was dangerous
  - B. Punishment helps learning
  - C. The pupils were honest
  - D. They were suited to teaching
  
2. The teacher-subjects were instructed to
  - A. Stop when pupil asked them to
  - B. Denounce pupils who made mistakes
  - C. Reduce the shock level after a correct answer
  - D. Give punishment according to the rule.
  
3. Before the experiment took place the psychiatrists
  - A. Believed that shock of 150 volts was too dangerous.
  - B. Failed to agree on how the teacher-subjects would respond to instructions.
  - C. Underestimated the teacher-subjects willingness to comply with experimental procedure.
  - D. Thought that many of the teacher-subjects would administer a sock of 450 volts.

## **Practise Reading Passage 1.2**

The concept of the rocket, or rather the mechanism behind the idea of propelling an object into the air, has been around for well over two thousand years. However, it wasn't until the discovery of the reaction principle, which was the key to space travel and so represents one of the great milestones in the history of scientific thought, that rocket technology was able to develop. Not only did it solve a problem that had intrigued man for ages, but, more importantly, it literally opened the door to exploration of the universe.

An intellectual breakthrough, brilliant though it may be, does not automatically ensure that the transition is made from theory to practice. Despite the fact that rockets had been used sporadically for several hundred years, they remained a relatively minor artefact of civilisation until the twentieth century. Prodigious efforts, accelerated during two world wars, were required before the technology of primitive rocketry could be translated into the reality of sophisticated astronauts. It is strange that the rocket was generally ignored by writers of fiction to transport their heroes to mysterious realms beyond the Earth, even though it had been commonly used in fireworks displays in China since the thirteenth century. The reason is that nobody associated the reaction principle with the idea of travelling through space to a neighbouring world.

A simple analogy can help us to understand how a rocket operates. It is much like a machine gun mounted on the rear of a boat. In reaction to the backward discharge of bullets, the gun, and hence the boat, move forwards. A rocket motor's 'bullets' are minute, high-speed particles produced by burning propellants in a suitable chamber. The reaction to the ejection of these small particles causes the rocket to move forwards. There is evidence that the reaction principle was applied practically well before the rocket was invented. In his *Noctes Atticae* or Greek Nights, Aulus Gellius describes 'the pigeon of Archytas', an invention dating back to about 360 BC. Cylindrical in shape, made of wood, and hanging from string, it was moved to and fro by steam blowing out from small exhaust ports at either end. The reaction to the discharging steam provided the bird with motive power.

The invention of rockets is linked inextricably with the invention of 'black powder'. Most historians of technology credit the Chinese with its discovery. They base their belief on studies of Chinese writings or on the notebooks of early Europeans who settled in or made long visits to China to study its history and civilisation. It is probable that, some time in the tenth century, black powder was first compounded from its basic ingredients of saltpetre, charcoal and sulphur. But this does not mean that it was immediately used to propel rockets. By the thirteenth century, powder-propelled fire arrows had become rather common. The Chinese relied on this type of technological development to produce incendiary projectiles of many sorts, explosive grenades and possibly cannons to repel their enemies. One such weapon was the 'basket of fire' or, as directly translated from Chinese, the 'arrows like flying leopards'. The 0.7 metre-long arrows, each with a long tube of gunpowder attached near the point of each arrow, could be fired from a long, octagonal-shaped basket at the same time and had a range of 400 paces. Another weapon was the 'arrow as a flying sabre', which could be fired from crossbows. The rocket, placed in a similar position to other rocket-propelled arrows, was designed to increase the range. A small

iron weight was attached to the 1.5m bamboo shaft, just below the feathers, to increase the arrow's stability by moving the centre of gravity to a position below the rocket. At a similar time, the Arabs had developed the 'egg which moves and burns'. This 'egg' was apparently full of gunpowder and stabilised by a 1.5m tail. It was fired using two rockets attached to either side of this tail.

It was not until the eighteenth century that Europe became seriously interested in the possibilities of using the rocket itself as a weapon of war and not just to propel other weapons. Prior to this, rockets were used only in pyrotechnic displays. The incentive for the more aggressive use of rockets came not from within the European continent but from far-away India, whose leaders had built up a corps of rocketeers and used rockets successfully against the British in the late eighteenth century. The Indian rockets used against the British were described by a British Captain serving in India as 'an iron envelope about 200 millimetres long and 40 millimetres in diameter with sharp points at the top and a 3m-long bamboo guiding stick'. In the early nineteenth century the British began to experiment with incendiary barrage rockets. The British rocket differed from the Indian version in that it was completely encased in a stout, iron cylinder, terminating in a conical head, measuring one metre in diameter and having a stick almost five metres long and constructed in such a way that it could be firmly attached to the body of the rocket. The Americans developed a rocket, complete with its own launcher, to use against the Mexicans in the mid-nineteenth century. A long cylindrical tube was propped up by two sticks and fastened to the top of the launcher, thereby allowing the rockets to be inserted and lit from the other end. However, the results were sometimes not that impressive as the behaviour of the rockets in flight was less than predictable.

Since then, there have been huge developments in rocket technology, often with devastating results in the forum of war. Nevertheless, the modern day space programs owe their success to the humble beginnings of those in previous centuries who developed the foundations of the reaction principle. Who knows what it will be like in the future?

**Choose the correct letter, A, B, C or D.** Write your answers next to the question number.

1. The greatest outcome of the discovery of the reaction principle was that
  - A. Rockets could be propelled into the air.
  - B. Space travel became a reality.
  - C. A major problem had been solved.
  - D. Bigger rockets were able to be built.
2. According to the text, the greatest progress in rocket technology was made
  - A. From the tenth to the thirteenth centuries.
  - B. From the seventeenth to the nineteenth centuries.
  - C. From the early nineteenth to the late nineteenth century.
  - D. From the late nineteenth century to the present day.

## **Practise Reading Passage 1.3**

### **THE RISKS OF CIGARETTE SMOKE**

Discovered in the early 1800s and named nicotianine, the oily essence now called nicotine is the main active ingredient of tobacco. Nicotine, however, is only a small component of cigarette smoke, which contains more than 4,700 chemical compounds, including 43 cancer-causing substances. In recent times, scientific research has been providing evidence that years of cigarette smoking vastly increases the risk of developing fatal medical conditions.

In addition to being responsible for more than 85 per cent of lung cancers, smoking is associated with cancers of, amongst others, the mouth, stomach and kidneys, and is thought to cause about 14 per cent of leukemia and cervical cancers. In 1990, smoking caused more than 84,000 deaths, mainly resulting from such problems as pneumonia, bronchitis and influenza. Smoking, it is believed, is responsible for 30 per cent of all deaths from cancer and clearly represents the most important preventable cause of cancer in countries like the United States today.

Passive smoking, the breathing in of the side-stream smoke from the burning of tobacco between puffs or of the smoke exhaled by a smoker, also causes a serious health risk. A report published in 1992 by the US Environmental Protection Agency (EPA) emphasized the health dangers, especially from side-stream smoke. This type of smoke contains more, smaller particles and is therefore more likely to be deposited deep in the lungs. On the basis of this report, the EPA has classified environmental tobacco smoke in the highest risk category for causing cancer.

As an illustration of the health risks, in the case of a married couple where one partner is a smoker and one a non-smoker, the latter is believed to have a 30 per cent higher risk of death from heart disease because of passive smoking. The risk of lung cancer also increases over the years of exposure and the figure jumps to 80 per cent if the spouse has been smoking four packs a day for 20 years. It has been calculated that 17 per cent of cases of lung cancer can be attributed to high levels of exposure to second-hand tobacco smoke during childhood and adolescence.

A more recent study by researchers at the University of California at San Francisco (UCSF) has shown that second-hand cigarette smoke does more harm to non-smokers than to smokers. Leaving aside the philosophical question of whether anyone should have to breathe someone else's cigarette smoke, the report suggests that the smoke experienced by many people in their daily lives is enough to produce substantial adverse effects on a person's heart and lungs.

The report, published in the Journal of the American Medical Association (AMA), was based on the researchers' own earlier research but also includes a review of studies over the past few years. The American Medical Association represents about half of all US doctors and is a strong opponent of smoking. The study suggests that people who smoke cigarettes are continually damaging their cardiovascular system, which adapts in order to compensate for the effects of smoking. It further states that people who do not smoke do not have the benefit of their system adapting to the smoke inhalation. Consequently, the effects of passive smoking are far greater on non-smokers than on smokers.

This report emphasizes that cancer is not caused by a single element in cigarette smoke; harmful effects to health are caused by many components. Carbon monoxide, for example, competes with oxygen in red blood cells and interferes with the blood's ability to deliver life giving oxygen to the heart. Nicotine and other toxins in cigarette smoke activate small blood cells called platelets, which increases the likelihood of blood clots, thereby affecting blood circulation throughout the body.

The researchers criticize the practice of some scientific consultants who work with the tobacco industry for assuming that cigarette smoke has the same impact on smokers as it does on non-smokers. They argue that those scientists are underestimating the damage done by passive smoking and, in support of their recent findings, cite some previous research which points to passive smoking as the cause for between 30,000 and 60,000 deaths from heart attacks each year in the United States. This means that passive smoking is the third most preventable cause of death after active smoking and alcohol-related diseases.

The study argues that the type of action needed against passive smoking should be similar to that being taken against illegal drugs and AIDS (SIDA). The UCSF researchers maintain that the simplest and most cost-effective action is to establish smoke-free work places, schools and public places.

**Choose the correct letter, A, B, C or D.** Write your answers next to the question number.

1. According to the information in the text, leukaemia and pneumonia
  - A. Are responsible for 84,000 deaths each year.
  - B. Are strongly linked to cigarette smoking.
  - C. Are strongly linked to lung cancer.
  - D. Result in 30 per cent of deaths per year.
  
2. According to the information in the text, intake of carbon monoxide
  - A. Inhibits the flow of oxygen to the heart.
  - B. Increases absorption of other smoke particles.
  - C. Inhibits red blood formation.
  - D. Promotes nicotine absorption.
  
3. According to the information in the text, intake of nicotine encourages
  - A. Blood circulation through the body.
  - B. Activity of other toxins in the blood.
  - C. Formation of blood clots.
  - D. An increase of platelets in the blood.

## **Practise Reading Passage 1.4**

### **Environmental Management**

#### **Section A**

The role of governments in environmental management is difficult but inescapable. Sometimes, the state tries to manage the resources it owns, and does so badly. Often, however, governments act in an even more harmful way. They actually subsidise the exploitation and consumption of natural resources. A whole range of policies, from farm- price support to protection for coal-mining, do environmental damage and (often) make no economic sense. Scrapping them offers a two-fold bonus: a cleaner environment and a more efficient economy. Growth and environmentalism can actually go hand in hand, if politicians have the courage to confront the vested interest that subsidies create.

#### **Section B**

No activity affects more of the earth's surface than farming. It shapes a third of the planet's land area, not counting Antarctica, and the proportion is rising. World food output per head has risen by 4 per cent between the 1970s and 1980s mainly as a result of increases in yields from land already in cultivation, but also because more land has been brought under the plough. Higher yields have been achieved by increased irrigation, better crop breeding, and a doubling in the use of pesticides and chemical fertilisers in the 1970s and 1980s.

#### **Section C**

All these activities may have damaging environmental impacts. For example, land clearing for agriculture is the largest single cause of deforestation; chemical fertilisers and pesticides may contaminate water supplies; more intensive farming and the abandonment of fallow periods tend to exacerbate soil erosion; and the spread of mono-Culture and use of high-yielding varieties of crops have been accompanied by the disappearance of old varieties of food plants which might have provided some insurance against pests or diseases in future. Soil erosion threatens the productivity of land in both rich and poor countries. The United States, where the most careful measurements have been done, discovered in 1982 that about one-fifth of its farmland is losing topsoil at a rate likely to diminish the soil's productivity. The country subsequently embarked upon a program to convert 11 per cent of its cropped land to meadow or forest. Topsoil in India and China is vanishing much faster than in America.

#### **Section D**

Government policies have frequently compounded the environmental damage that farming can cause. In the rich countries, subsidies for growing crops and price supports for farm output drive up the price of land. The annual value of these subsidies is immense: about \$250 billion, or more than all World Bank lending in the 1980s. To increase the output of crops per acre, a farmer's easiest option is to use more of the most readily available inputs: fertilisers and pesticides. Fertiliser use doubled in Denmark in the period 1960-1985 and increased in The Netherlands by 150 per cent. The quantity of pesticides applied has risen too; by 69 per cent in 1975-1984 in Denmark, for example, with a rise of 115 per cent in the frequency of application in the three years from 1981.

In the late 1980s and early 1990s some efforts were made to reduce farm subsidies. The most dramatic example was that of New Zealand, which scrapped most farm support in 1984. A study of the environmental effects, conducted in 1993, found that the end of fertiliser subsidies had been followed by a fall in fertiliser use (a fall compounded by the decline in world commodity prices, which cut farm incomes). The removal of subsidies also stopped land-clearing and overstocking, which in the past had been the principal causes of erosion. Farms began to diversify. The one kind of subsidy whose removal appeared to have been bad for the environment was the subsidy to manage soil erosion,

In less enlightened countries, and in the European Union, the trend has been to reduce rather than eliminate subsidies, and to introduce new payments to encourage farmers to treat their land in environmentally friendlier ways, or to leave it fallow. It may sound strange but such payments need to be higher than the existing incentives for farmers to grow food crops. Farmers, however, dislike being paid to do nothing. In several countries they have become interested in the possibility of using fuel produced from crop residues either as a replacement for petrol (as ethanol) or as fuel for power stations (as biomass). Such fuels produce far less carbon dioxide than coal or oil, and absorb carbon dioxide as they grow. They are therefore less likely to contribute to the greenhouse effect. But they are rarely competitive with fossil fuels unless subsidised - and growing them does no less environmental harm than other crops.

#### Section E

In poor countries, governments aggravate other sorts of damage. Subsidies for pesticides and artificial fertilisers encourage farmers to use greater quantities than are needed to get the highest economic crop yield. A study by the International Rice Research Institute of pesticide use by farmers in South East Asia found that, with pest-resistant varieties of rice, even moderate applications of pesticide frequently cost farmers more than they saved. Such waste puts farmers on a chemical treadmill: bugs and weeds become resistant to poisons, so next year's poisons must be more lethal. One cost is to human health. Every year some 10,000 people die from pesticide poisoning, almost all of them in the developing countries, and another 400,000 become seriously ill. As for artificial fertilisers, their use world-wide increased by 40 per cent per unit of farmed land between the mid 1970s and late 1980s, mostly in the developing countries. Overuse of fertilisers may cause farmers to stop rotating crops or leaving their land fallow. That, in turn, may make soil erosion worse.

#### Section F

A result of the Uruguay Round of world trade negotiations is likely to be a reduction of 36 per cent in the average levels of farm subsidies paid by the rich countries in 1986-1990. Some of the world's food production will move from Western Europe to regions where subsidies are lower or non-existent, such as the former communist countries and parts of the developing world. Some environmentalists worry about this outcome. It will undoubtedly mean more pressure to convert natural habitat into farmland. But it will also have many desirable environmental effects. The intensity of farming in the rich world should decline, and the use of chemical inputs will diminish. Crops are more likely to be grown in the environments to which they are naturally suited. And more farmers in poor countries will have the money and the incentive to manage

their land in ways that are sustainable in the long run. That is important. To feed an increasingly hungry world, farmers need every incentive to use their soil and water effectively and efficiently.

**Choose the correct letter, A, B, C D or E.** Write your answers next to the question number.

1. Research completed in 1982 found that in the United States soil erosion
  - A. Reduced productivity of farmland by 20 per cent.
  - B. Was almost as severe as in India and China.
  - C. Was causing significant damage to 20 per cent of farmland.
  - D. Could be reduced by converting cultivated land to meadow or forest.
  
2. By mid-1980's, farmers in Denmark
  - A. Used 50 per cent less fertiliser than Dutch farmers.
  - B. Used twice as much fertiliser as they had in 1960.
  - C. Applied fertiliser much more frequently than in 1960.
  - D. More than doubled the amount of pesticide they used in just 3 years.
  
3. Which of the following increased in New Zealand after 1984?
  - A. Farm incomes
  - B. Use of fertiliser
  - C. Over-stocking
  - D. Farm diversification
  
4. The writer refers to some rich countries as being less 'less enlightened' than New Zealand because
  - A. They disapprove of paying farmers for not cultivating the land.
  - B. Their new fuel crops are as harmful as the ones they have replaced.
  - C. Their policies do not recognise the long-term benefit of ending subsidies.
  - D. They have encouraged their farmers to follow environmentally friendly practices.
  
5. The writer believes that the Uruguay Round agreements on trade will
  - A. Encourage more sustainable farming practices in the long term.
  - B. Do more harm than good to the international environment.
  - C. Increase pressure to cultivate land in rich countries.
  - D. Be more beneficial to rich than to poor countries.
  
6. The most suitable title for the above reading passage is
  - A. Environmental management
  - B. Increasing the world's food supply
  - C. Soil erosion
  - D. Fertilisers and pesticides – the way forward
  - E. Farm subsidies

## **Practise Reading Passage 1.5**

### **SECRETS OF THE FOREST**

- A. In 1942 Allan R Holmberg, a doctoral student in anthropology from Yale University, USA, ventured deep into the jungle of Bolivian Amazonia and searched out an isolated band of Siriono Indians. The Siriono, Holmberg later wrote, led a "strikingly backward" existence. Their villages were little more than clusters of thatched huts. Life itself was a perpetual and punishing search for food: some families grew manioc and other starchy crops in small garden plots cleared from the forest, while other members of the tribe scoured the country for small game and promising fish holes. When local resources became depleted, the tribe moved on.

As for technology, Holmberg noted, the Siriono "may be classified among the most handicapped peoples of the world". Other than bows, arrows and crude digging sticks, the only tools the Siriono seemed to possess were "two machetes worn to the size of pocket-knives".

- B. Although the lives of the Siriono have changed in the intervening decades, the image of them as Stone Age relics has endured. Indeed, in many respects the Siriono epitomize the popular conception of life in Amazonia. To casual observers, as well as to influential natural scientists and regional planners, the luxuriant forests of Amazonia seem ageless, unconquerable, a habitat totally hostile to human civilization. The apparent simplicity of Indian ways of life has been judged an evolutionary adaptation to forest ecology, living proof that Amazonia could not - and cannot - sustain a more complex society. Archaeological traces of far more elaborate cultures have been dismissed as the ruins of invaders from outside the region, abandoned to decay in the uncompromising tropical environment.
- C. The popular conception of Amazonia and its native residents would be enormously consequential if it were true. But the human history of Amazonia in the past 11,000 years betrays that view as myth. Evidence gathered in recent years from anthropology and archaeology indicates that the region has supported a series of indigenous cultures for eleven thousand years; an extensive network of complex societies - some with populations perhaps as large as 100,000 - thrived there for more than 1,000 years before the arrival of Europeans. (Indeed, some contemporary tribes, including the Siriono, still live among the earthworks of earlier cultures.) Far from being evolutionarily retarded, prehistoric Amazonian people developed technologies and cultures that were advanced for their time. If the lives of Indians today seem "primitive", the appearance is not the result of some environmental adaptation or ecological barrier; rather it is a comparatively recent adaptation to centuries of economic and political pressure. Investigators who argue otherwise have unwittingly projected the present onto the past.
- D. The evidence for a revised view of Amazonia will take many people by surprise. Ecologists have assumed that tropical ecosystems were shaped entirely by natural forces and they have

focused their research on habitats they believe have escaped human influence. But as the University of Florida ecologist, Peter Feinsinger, has noted, an approach that leaves people out of the equation is no longer tenable. The archaeological evidence shows that the natural history of Amazonia is to a surprising extent tied to the activities of its prehistoric inhabitants.

- E. The realization comes none too soon. In June 1992 political and environmental leaders from across the world met in Rio de Janeiro to discuss how developing countries can advance their economies without destroying their natural resources. The challenge is especially difficult in Amazonia. Because the tropical forest has been depicted as ecologically unfit for large-scale human occupation, some environmentalists have opposed development of any kind.

Ironically, one major casualty of that extreme position has been the environment itself. While policy makers struggle to define and implement appropriate legislation, development of the most destructive kind has continued apace over vast areas.

- F. The other major casualty of the "naturalism" of environmental scientists has been the indigenous Amazonians, whose habits of hunting, fishing, and slash-and-burn cultivation often have been represented as harmful to the habitat. In the clash between environmentalists and developers, the Indians, whose presence is in fact crucial to the survival of the forest, have suffered the most. The new understanding of the pre-history of Amazonia, however, points toward a middle ground. Archaeology makes clear that with judicious management selected parts of the region could support more people than anyone thought before. The long-buried past, it seems, offers hope for the future.

**Choose the correct letter, A, B, C or D. Write your answers next to the question number.**

1. In 1942 the US anthropology student concluded that the Siriono
  - A. Were unusually aggressive and cruel.
  - B. Had their way of life destroyed by invaders.
  - C. Were an extremely primitive society.
  - D. Had only recently made permanent settlements.
  
2. The author believes recent discoveries of the remains of complex societies in Amazonia
  - A. Are evidence of early indigenous communities
  - B. Are the remains of settlements by invaders
  - C. Are the ruins of communities established since the European invasions.
  - D. Show the region has only relatively recently been covered by forest.
  
3. The assumption that the tropical ecosystem of Amazonia has been created solely by natural forces
  - A. Has often been questioned by ecologists in the past.
  - B. Has been shown to be incorrect by recent research.
  - C. Was made by Peter Feinsinger and other ecologists.
  - D. Has led to some fruitful discoveries.
  
4. The application of our new insights into the Amazonian past would be
  - A. Warn us against allowing any development at all.
  - B. Cause further suffering to the Indian communities.
  - C. Change present policies on development in the region.
  - D. Reduce the amount of hunting, fishing, and ‘slash-and-burn’.

## Practise Reading Passage 1.6

### HIGHS & LOWS



Hormone levels - and hence our moods - may be affected by the weather. Gloomy weather can cause depression, but sunshine appears to raise the spirits. In Britain, for example, the dull weather of winter drastically cuts down the amount of sunlight that is experienced which strongly affects some people. They become so depressed and lacking in energy that their work and social life are affected. This condition has been given the name SAD (Seasonal Affective Disorder). Sufferers can fight back by making the most of any sunlight in winter and by spending a few hours each day under special, full-spectrum lamps. These provide more ultraviolet and blue-green light than ordinary fluorescent and tungsten lights. Some Russian scientists claim that children learn better after being exposed to ultraviolet light. In warm countries, hours of work are often arranged so that workers can take a break, or even a siesta, during the hottest part of the day. Scientists are working to discover the links between the weather and human beings' moods and performance.

It is generally believed that tempers grow shorter in hot, muggy weather. There is no doubt that 'crimes against the person' rise in the summer, when the weather is hotter and

fall in the winter when the weather is colder. Research in the United States has shown a relationship between temperature and street riots. The frequency of riots rises dramatically as the weather gets warmer, hitting a peak around 27-30°C. But is this effect really due to a mood change caused by the heat? Some scientists argue that trouble starts more often in hot weather merely because there are more people in the street when the weather is good.

Psychologists have also studied how being cold affects performance. Researchers compared divers working in icy cold water at 5°C with others in water at 20°C (about swimming pool temperature). The colder water made the divers worse at simple arithmetic and other mental tasks. But significantly, their performance was impaired as soon as they were put into the cold water - before their bodies had time to cool down. This suggests that the low temperature did not slow down mental functioning directly, but the feeling of cold distracted the divers from their tasks.

Psychologists have conducted studies showing that people become less sceptical and more optimistic when the weather is sunny. However, this apparently does not just depend on the temperature. An

American psychologist studied customers in a temperature-controlled restaurant. They gave bigger tips when the sun was shining and smaller tips when it wasn't, even though the temperature in the restaurant was the same. A link between weather and mood is made believable by the evidence for a connection between behaviour and the length of the daylight hours. This in turn might involve the level of a hormone called melatonin, produced in the pineal gland in the brain. The amount of melatonin falls with greater exposure to daylight. Research shows that melatonin plays an important part in the seasonal behaviour of certain animals. For example, food consumption of stags increases during the winter, reaching a peak in February/ March. It falls again to a low point in May, then rises to a peak in September, before dropping to another minimum in November. These changes seem to be triggered by varying melatonin levels.

In the laboratory, hamsters put on more weight when the nights are getting shorter and their melatonin levels are falling. On the other hand, if they are given injections of melatonin, they will stop eating altogether. It seems that time cues provided by the changing lengths of day and night trigger changes in animals' behaviour - changes that are needed to cope with the cycle of the seasons. People's moods too, have been shown to react to the length of the daylight hours. Sceptics might say that longer

exposure to sunshine puts people in a better mood because they associate it with the happy feelings of holidays and freedom from responsibility. However, the belief that rain and murky weather make people more unhappy is borne out by a study in Belgium, which showed that a telephone counselling service gets more telephone calls from people with suicidal feelings when it rains.

When there is a thunderstorm brewing, some people complain of the air being 'heavy' and of feeling irritable, moody and on edge. They may be reacting to the fact that the air can become slightly positively charged when large thunderclouds are generating the intense electrical fields that cause lightning flashes. The positive charge increases the levels of serotonin (a chemical involved in sending signals in the nervous system). High levels of serotonin in certain areas of the nervous system make people more active and reactive and, possibly, more aggressive. When certain winds are blowing, such as the Mistral in southern France and the Fohn in southern Germany, mood can be affected - and the number of traffic accidents rises. It may be significant that the concentration of positively charged particles is greater than normal in these winds. In the United Kingdom, 400,000 ionizers are sold every year. These small machines raise the number of negative ions in the air in a room. Many people claim they feel better in negatively charged air.

**Choose the correct letter, A, B, C or D.** Write your answers next to the question number.

1. Why did the divers perform less well in colder conditions?
  - A. They were less able to concentrate.
  - B. Their body temperature fell too quickly.
  - C. Their mental functions were immediately affected by the cold.
  - D. They were used to swimming pool conditions.
  
2. The number of daylight hours
  - A. Affects the performance of workers in restaurants.
  - B. Influences animal feeding habits.
  - C. Makes animal like hamsters more active.
  - D. Prepares humans for having greater leisure time.
  
3. Human irritability may be influenced by
  - A. How nervous and aggressive people are.
  - B. Reaction to certain weather phenomenon.
  - C. The number of ions being generated by machines.
  - D. The attitude of people to thunderstorms.

## **Practise Reading Passage 1.7**

### **AIR POLLUTION**

#### **Part One**

- A. Air pollution is increasingly becoming the focus of government and citizen concern around the globe. From Mexico City and New York, to Singapore and Tokyo, new solutions to this old problem are being proposed, Mailed and implemented with ever increasing speed. It is feared that unless pollution reduction measures are able to keep pace with the continued pressures of urban growth, air quality in many of the world's major cities will deteriorate beyond reason.
- B. Action is being taken along several fronts: through new legislation, improved enforcement and innovative technology. In Los Angeles, state regulations are forcing manufacturers to try to sell ever cleaner cars: their first of the cleanest, titled "Zero Emission Vehicles", hove to be available soon, since they are intended to make up 2 per cent of sales in 1997. Local authorities in London are campaigning to be allowed to enforce anti-pollution lows themselves; at present only the police have the power to do so, but they tend to be busy elsewhere. In Singapore, renting out toad space to users is the way of the future.
- C. When Britain's Royal Automobile Club monitored the exhausts of 60,000 vehicles, it found that 12 per cent of them produced more than half the total pollution. Older cars were the worst offenders; though a sizeable number of quire new cars were also identified as gross polluters, they were simply badly tuned. California has developed a scheme to get these gross polluters off the streets: they offer a flat \$700 for any old, run-down vehicle driven in by its owner. The aim is to remove the heaviest-polluting, most decrepit vehicles from the roads.
- D. As part of a European Union environmental programme, a London council is resting an infra-red spectrometer from the University of Denver in Colorado. It gauges the pollution from a passing vehicle - more useful than the annual stationary rest that is the British standard today - by bouncing a beam through the exhaust and measuring what gets blocked. The councils next step may be to link the system to a computerised video camera able to read number plates automatically.
- E. The effort to clean up cars may do little to cut pollution if nothing is done about the tendency to drive them more. Los Angeles has some of the world's cleanest cars - far better than those of Europe - but the total number of miles those cars drive continues to grow. One solution is car-pooling, an arrangement in which a number of people who share the same destination share the use of one car. However, the average number of people in o car on the freeway in , which is 1.0, has been falling steadily. Increasing it would be an effective way of reducing emissions as well as easing congestion. The trouble is, Los Angeles seem to like being alone in their cars.

- F. Singapore has for a while had a scheme that forces drivers to buy a badge if they wish to visit a certain part of the city. Electronic innovations make possible increasing sophistication: rates can vary according to road conditions, time of day and so on. Singapore is advancing in this direction, with a city-wide network of transmitters to collect information and charge drivers as they pass certain points. Such road-pricing, however, can be controversial. When the local government in Cambridge, England, considered introducing Singaporean techniques, it faced vocal and ultimately successful opposition.

### **Part Two**

The scope of the problem facing the world's cities is immense. In 1992, the United Nations Environmental Programme and the World Health Organisation (WHO) concluded that all of a sample of twenty megacities - places likely to have more than ten million inhabitants in the year 2000 - already exceeded the level the WHO deems healthy in at least one major pollutant. Two-thirds of them exceeded the guidelines for two, seven for three or more.

Of the six pollutants monitored by the WHO - carbon dioxide, nitrogen dioxide, ozone, sulphur dioxide, lead and particulate matter - it is this last category that is attracting the most attention from health researchers. PM10, a sub-category of particulate matter measuring ten-millionths of a metre across, has been implicated in thousands of deaths a year in Britain alone. Research being conducted in two counties of Southern California is reaching similarly disturbing conclusions concerning this little-understood pollutant.

A world-wide rise in allergies, particularly asthma, over the past four decades is now said to be linked with increased air pollution. The lungs and brains of children who grow up in polluted air offer further evidence of its destructive power. The old and ill, however, are the most vulnerable to the acute effects of heavily polluted stagnant air. It can actually hasten death, as it did in December 1991 when a cloud of exhaust fumes lingered over the city of London for over a week.

The United Nations has estimated that in the year 2000 there will be twenty-four mega-cities and a further eighty-five cities of more than three million people. The pressure on public officials, corporations and urban citizens to reverse established trends in air pollution is likely to grow in proportion with the growth of cities themselves. Progress is being made. The question, though, remains the same: 'Will change happen quickly enough?'

**Choose the correct letter, A, B, C or D.** Write your answers next to the question number.

- A. How many pollutants currently exceed WHO guidelines in all megacities studies.
- A. One
  - B. Two
  - C. Three
  - D. Seven
- B. Which pollutant is currently the subject of urgent research?
- A. Nitrogen dioxide
  - B. Ozone
  - C. Lead
  - D. Particulate matter
- C. Which of the following groups of people are the most severely affected by intense air pollution?
- A. Allergy sufferers
  - B. Children
  - C. The old and ill
  - D. Asthma sufferers

## Practise Reading Passage 1.8

### VOTE FOR WOMEN

The suffragette movement, which campaigned for votes for women in the early twentieth century, is most commonly associated with the Pankhurst family and militant acts of varying degrees of violence. The Museum of London has drawn on its archive collection to convey a fresh picture with its exhibition.

The Purple, White and Green: Suffragettes in London 1906-14.

The name is a reference to the colour scheme that the Women's Social and Political Union (WSPU) created to give the movement a uniform, nationwide image. By doing so, it became one of the first groups to project a corporate identity, and it is this advanced marketing strategy, along with the other organisational and commercial achievements of the WSPU, to which the exhibition is devoted.

Formed in 1903 by the political campaigner Mrs Emmeline Pankhurst and her daughters Christabel and Sylvia, the WSPU began an educated campaign to put women's suffrage on the political agenda. New Zealand, Australia and parts of the United States had already enfranchised women, and growing numbers of their British counterparts wanted the same opportunity.

With their slogan 'Deeds not words', and the introduction of the colour scheme, the WSPU soon brought the movement the cohesion and focus it had previously lacked.

Membership grew rapidly as women deserted the many other, less directed, groups and joined it. By 1906 the WSPU headquarters, called the Women's Press Shop, had been established in Charing Cross Road and in spite of limited communications (no radio or television, and minimal use of the telephone) the message had spread around the country, with members and branch officers stretching to as far away as Scotland.

The newspapers produced by the WSPU, first Votes for Women and later The Suffragette, played a vital role in this communication. Both were sold throughout the country and proved an invaluable way of informing members of meetings, marches, fund-raising events and the latest news and views on the movement.

Equally importantly for a rising political group, the newspaper returned a profit. This was partly because advertising space was bought in the paper by large department stores such as Selfridges, and jewellers such as Mappin & Webb. These two, together with other like-minded commercial enterprises sympathetic to the cause, had quickly identified a direct way to reach a huge market of women, many with money to spend.

The creation of the colour scheme provided another money-making opportunity which the WSPU was quick to exploit. The group began to sell playing cards, board games, Christmas and greeting cards, and countless other goods, all in the purple, white and green colours. In 1906 such merchandising of a corporate identity was a new marketing concept.

But the paper and merchandising activities alone did not provide sufficient funds for the WSPU to meet organisational costs, so numerous other fund-raising activities combined to fill the coffers of the ‘war chest’. The most notable of these was the Woman’s Exhibition, which took place in 1909 in a Knightsbridge ice-skating rink, and in 10 days raised the equivalent of £250,000 today.

The Museum of London’s exhibition is largely visual, with a huge number of items on show. Against a quiet background hum of street sounds, copies of *The Suffragette*, campaign banners and photographs are all on display, together with one of Mrs shoes and a number of purple, white and green trinkets.

Photographs depict vivid scenes of a suffragette’s life: WSPU members on a self-proclaimed ‘monster’ march, wearing their official uniforms of a white frock decorated with purple, white and green accessories; women selling *The Suffragette* at street corners, or chalking up pavements with details of a forthcoming meeting.

Windows display postcards and greeting cards designed by women artists for the movement, and the quality of the artwork indicates the wealth of resources the WSPU could call on from its talented members.

Visitors can watch a short film made up of old newsreels and cinema material which clearly reveals the political mood of the day towards the suffragettes. The programme begins with a short film devised by the ‘antis’ - those opposed to women having the vote - depicting a suffragette as a fierce harriidan bullying her poor, abused husband.

Original newsreel footage shows the suffragette Emily Wilding Davison throwing herself under King George V’s horse at a famous race-

Although the exhibition officially charts the years 1906 to 1914, graphic display boards outlining the bills of enfranchisement of 1918 and 1928, which gave the adult female populace of Britain the vote, show what was achieved. It demonstrates how advanced the suffragettes were in their thinking, in the marketing of their campaign, and in their work as shrewd and skilful image-builders. It also conveys a sense of the energy and ability the suffragettes brought to their fight for freedom and equality. And it illustrates the intelligence employed by women who were at that time deemed by several politicians to have ‘brains too small to know how to vote’.

**Choose the correct letter, A, B, C D or E.** Write your answers next to the question number.

1. What is the main aspect of the suffragette movement's work to which the exhibition at the Museum of London devoted?
  - A. The role of Pankhurst family in the suffrage movement.
  - B. The violence of the movement's political campaign
  - C. The success of the movement's corporate image
  - D. The movement's co-operation with the suffrage groups overseas.
  
2. Why was the WSPU more successful than other suffrage groups?
  - A. Its leaders were much better educated.
  - B. It received funding from movements abroad.
  - C. It had access to new technology.
  - D. It had a clear purpose and direction.
  
3. In which **two** of the following years were laws passed allowing British women to vote?  
(Choose two letters here)
  - A. 1906
  - B. 1909
  - C. 1914
  - D. 1918
  - E. 1928
  
4. The writer of the article finds exhibition to be
  - A. Misleading
  - B. Exceptional
  - C. Disappointing
  - D. Informative

## Practise Reading Passage 1.9

### THE RETURN OF ARTIFICIAL INTELLIGENCE

*It is becoming acceptable again to talk of computers performing human tasks such as problem-solving and pattern-recognition.*

- A. After years in the wilderness, the term 'artificial intelligence' (AI) seems poised to make a comeback. AI was big in the 1980s but vanished in the 1990s. It re-entered public consciousness with the release of AI, a movie about a robot boy. This has ignited public debate about AI, but the term is also being used once more within the computer industry. Researchers, executives and marketing people are now using the expression without irony or inverted commas. And it is not always hype. The term is being applied, with some justification, to products that depend on technology that was originally developed by AI researchers. Admittedly, the rehabilitation of the term has a long way to go, and some firms still prefer to avoid using it. But the fact that others are starting to use it again suggests that AI has moved on from being seen as an over-ambitious and under-achieving field of research.
- B. The field was launched, and the term 'artificial intelligence' coined, at a conference in 1956, by a group of researchers that included Marvin Minsky, John McCarthy, Herbert Simon and Alan Newell, all of whom went on to become leading figures in the field. The expression provided an attractive but informative name for a research programme that encompassed such previously disparate fields as operations research, cybernetics, logic and computer science. The goal they shared was an attempt to capture or mimic human abilities using machines. That said, different groups of researchers attacked different problems, from speech recognition to chess playing, in different ways; AI unified the field in name only. But it was a term that captured the public imagination.
- C. Most researchers agree that AI peaked around 1985. A public reared on science-fiction movies and excited by the growing power of computers had high expectations. For years, AI researchers had implied that a breakthrough was just around the corner. Marvin Minsky said in 1967 that within a generation the problem of creating 'artificial intelligence' would be substantially solved. Prototypes of medical-diagnosis programs and speech recognition software appeared to be making progress. It proved to be a false dawn. Thinking computers and household robots failed to materialise, and a backlash ensued. 'There was undue optimism in the early 1980s,' says David Leake, a researcher at Indiana University. 'Then when people realised these were hard problems, there was retrenchment. By the late 1980s, the term AI was being avoided by many researchers, who opted instead to align themselves with specific sub-disciplines such as neural networks, agent technology, case-based reasoning, and so on.'
- D. Ironically, in some ways AI was a victim of its own success. Whenever an apparently mundane problem was solved, such as building a system that could land an aircraft

unattended, the problem was deemed not to have been AI in the first place. 'If it works, it can't be AI,' as Dr Leake characterises it. The effect of repeatedly moving the goal-posts in this way was that AI came to refer to 'blue-sky' research that was still years away from commercialisation. Researchers joked that AI stood for 'almost implemented'. Meanwhile, the technologies that made it once the market, such as speech recognition, language translation and decision-support software, were no longer regarded as AI. Yet all three once fell well within the umbrella of AI research.

- E. But the tide may now be turning, according to Dr Leake. HNC Software of San Diego, backed by a government agency, reckon that their new approach to artificial intelligence is the most powerful and promising approach ever discovered. HNC claim that their system, based on a duster of 30 processors, could be used to spot camouflaged vehicles on a battlefield or extract a voice signal from a noisy background - tasks humans can do well, but computers cannot. 'Whether or not their technology lives up to the claims made for it, the fact that HNC are emphasising the use of AI is itself an interesting development,' says Dr Leake.
- F. Another factor that may boost the prospects for AI in the near future is that investors are now looking for firms using clever technology, rather than just a clever business model, to differentiate themselves. In particular, the problem of information overload, exacerbated by the growth of e-mail and the explosion in the number of web pages, means there are plenty of opportunities for new technologies to help filter and categorise information - classic AI problems. That may mean that more artificial intelligence companies will start to emerge to meet this challenge.
- G. The 1969 film, 2001:A Space Odyssey, featured an intelligent computer called HAL 9000. As well as understanding and speaking English, HAL could play chess and even learned to lipread. HAL thus encapsulated the optimism of the 1960s that intelligent computers would be widespread by 2001. But 2001 has been and gone, and there is still no sign of a HAL-like computer. Individual systems can play chess or transcribe speech, but a general theory of machine intelligence still remains elusive. It may be, however, that the comparison with HAL no longer seems quite so Important, and AI can now be judged by what it can do, rather than by how well it matches up to a 30-year-old science-fiction film. 'People are beginning to realise that there are impressive things that these systems can do,' says Dr Leake hopefully.

**Choose the correct letter, A, B, C or D.** Write your answers next to the question number.

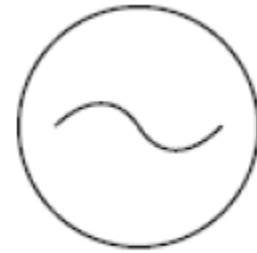
1. According to researchers, in the late 1980s there was a feeling that
  - A. A general theory of AI would never be developed.
  - B. Original expectations of AI may not have been justified.
  - C. A wide range of applications was close to fruition.
  - D. More powerful computers were the key to further progress.
  
2. In Dr. Leake's opinion, the reputation of AI suffered as result of
  - A. Changing perceptions.
  - B. Premature implementation.
  - C. Poorly planned projects.
  - D. Commercial pressures.
  
3. The prospects for AI may benefit from
  - A. Existing AI applications.
  - B. New business models.
  - C. Orders from internet-only companies.
  - D. New investment priorities.

## Practise Reading Passage 1.10

### VISUAL SYMBOLS AND THE BLIND

#### Part 1

From a number of recent studies, it has become clear that blind people can appreciate the use of outlines and perspectives to describe the arrangement of objects and other surfaces in space. But pictures are more than literal representations. This fact was drawn to my attention dramatically when a blind woman in one of my investigations decided on her own initiative to draw a wheel as it was spinning. To show this motion, she traced a curve inside the circle (Fig. 1). I was taken aback. Lines of motion, such as the one she used, are a very recent invention in the history of illustration. Indeed, as art scholar David Kunzle notes, Wilhelm Busch, a trend-setting nineteenth-century cartoonist, used virtually no motion lines in his popular figures until about 1877. Fig. 1



**Fig. 1**

When I asked several other blind study subjects to draw a spinning wheel, one particularly clever rendition appeared repeatedly: several subjects showed the wheel's spokes as curved lines. When asked about these curves, they all described them as metaphorical ways of suggesting motion. Majority rule would argue that this device somehow indicated motion very well. But was it a better indicator than, say, broken or wavy lines - or any other kind of line, for that matter? The answer was not clear. So I decided to test whether various lines of motion were apt ways of showing movement or if they were merely idiosyncratic marks. Moreover, I wanted to discover whether there were differences in how the blind and the sighted interpreted lines of motion.

To search out these answers, I created raised-line drawings of five different wheels, depicting spokes with lines that curved, bent, waved, dashed and extended beyond the perimeter of the wheel. I then asked eighteen blind volunteers to feel the wheels and assign one of the following motions to each wheel: wobbling, spinning fast, spinning steadily, jerking or braking. My control group consisted of eighteen sighted undergraduates from the University of Toronto.

All but one of the blind subjects assigned distinctive motions to each wheel. Most guessed that the curved spokes indicated that the wheel was spinning steadily; the wavy spokes, they thought, suggested that the wheel was wobbling; and the bent spokes were taken as a sign that the wheel was jerking. Subjects assumed that spokes extending beyond the wheel's perimeter signified that the wheel had its brakes on and that dashed spokes indicated the wheel was spinning quickly.

In addition, the favoured description for the sighted was the favoured description for the blind in every instance. What is more, the consensus among the sighted was barely higher than that among the blind. Because motion devices are unfamiliar to the blind, the task I gave them involved some problem solving. Evidently, however, the blind not only figured out meanings for each line of motion, but as a group they generally came up with the same meaning at least as frequently as did sighted subjects.

## Part 2

We have found that the blind understand other kinds of visual metaphors as well. One blind woman drew a picture of a child inside a heart - choosing that symbol, she said, to show that love surrounded the child. With Chang Hong Liu, a doctoral student from China, I have begun exploring how well blind people understand the symbolism behind shapes such as hearts that do not directly represent their meaning.

We gave a list of twenty pairs of words to sighted subjects and asked them to pick from each pair the term that best related to a circle and the term that best related to a square. For example, we asked: What goes with soft? A circle or a square? Which shape goes with hard?

All our subjects deemed the circle soft and the square hard. A full 94% ascribed happy to the circle, instead of sad. But other pairs revealed less agreement: 79% matched fast to slow and weak to strong, respectively. And only 51% linked deep to circle and shallow to square. (See Fig. 2.) When we tested four totally blind volunteers using the same list, we found that their choices closely resembled those made by the sighted subjects. One man, who had been blind since birth, scored extremely well. He made only one match differing from the consensus, assigning 'far' to square and 'near' to circle. In fact, only a small majority of sighted subjects - 53% - had paired far and near to the opposite partners. Thus, we concluded that the blind interpret abstract shapes as sighted people do.

| Words associated with circle/square | Agreement among subjects (%) |
|-------------------------------------|------------------------------|
| <b>SOFT-HARD</b>                    | <b>100</b>                   |
| <b>MOTHER-FATHER</b>                | <b>94</b>                    |
| <b>HAPPY-SAD</b>                    | <b>94</b>                    |
| <b>GOOD-EVIL</b>                    | <b>89</b>                    |
| <b>LOVE-HATE</b>                    | <b>89</b>                    |
| <b>ALIVE-DEAD</b>                   | <b>87</b>                    |
| <b>BRIGHT-DARK</b>                  | <b>87</b>                    |
| <b>LIGHT-HEAVY</b>                  | <b>85</b>                    |
| <b>WARM-COLD</b>                    | <b>81</b>                    |
| <b>SUMMER-WINTER</b>                | <b>81</b>                    |
| <b>WEAK-STRONG</b>                  | <b>79</b>                    |
| <b>FAST-SLOW</b>                    | <b>79</b>                    |
| <b>CAT-DOG</b>                      | <b>74</b>                    |
| <b>SPRING-FALL</b>                  | <b>74</b>                    |
| <b>QUIET-LOUD</b>                   | <b>62</b>                    |
| <b>WALKING-STANDING</b>             | <b>62</b>                    |
| <b>ODD-EVEN</b>                     | <b>57</b>                    |
| <b>FAR-NEAR</b>                     | <b>53</b>                    |
| <b>PLANT-ANIMAL</b>                 | <b>53</b>                    |
| <b>DEEP-SHALLOW</b>                 | <b>51</b>                    |

Fig. 2 Subjects were asked which word in each pair fits best with a circle and which with a square. These percentages show the level of consensus among sighted subjects.

**Choose the correct letter, A, B, C or D.** Write your answers next to the question number.

1. In the first paragraph the writer makes the point that blind people
  - A. May be interested in studying art.
  - B. Can draw outlines of different objects and surfaces.
  - C. Can recognise conventions such as perspective.
  - D. Can draw accurately.
2. The writer was surprised because the blind woman
  - A. Drew a circle on her own initiative.
  - B. Did not understand what a wheel looked like.
  - C. Included a symbol representing movement.
  - D. Was the first person to use lines of motion.
3. From the experiment described in part 1, the writer found that the blind subjects
  - A. Had good understanding of symbols representing movement.
  - B. Could control the movement of wheels very accurately.
  - C. Worked together well as a group in solving problems.
  - D. Got better than the sighted undergraduates.
4. Which of the following statements best summarises the writer's general conclusion?
  - A. The blind represent some aspects of reality differently from sighted people.
  - B. The blind comprehend visual metaphors in similar ways to sighted people.
  - C. The blind may create unusual and effective symbols to represent reality.
  - D. The blind may be successful artists if given the right training.

## **ANSWER KEY**

### **PASSAGE 1.1**

1. B
2. D
3. C

### **PASSAGE 1.2**

1. B
2. D

### **PASSAGE 1.3**

1. B
2. A
3. C

### **PASSAGE 1.4**

1. C
2. B
3. D
4. C
5. A
6. A

### **PASSAGE 1.5**

1. C
2. A
3. B
4. C

### **PASSAGE 1.6**

1. A
2. B
3. B

### **PASSAGE 1.7**

1. A
2. D
3. C

### **PASSAGE 1.8**

1. C
2. D
3. D and E
4. D

### **PASSAGE 1.9**

1. B
2. A
3. D

### **PASSAGE 1.10**

1. C
2. C
3. A
4. B